UET NOWSHERA | ELECTRICAL ENGINEERING

LAB REPORT 05

THE WHILE AND DO...WHILE LOOPS

1. OBJECTIVE

- To understand the syntax and execution flow of the while loop.
- To implement and execute a do...while loop.
- To utilize nested while loops for complex iterations.
- To differentiate between pre-test and post-test loop structures.

2. THEORETICAL BACKGROUND

The while Loop. A while loop is a control flow statement that allows code to be executed repeatedly based on a given boolean condition. The condition is evaluated first; if it is true, the code within the block is executed. This process repeats until the condition evaluates to false.

Nested while Loop. A nested while loop is simply a while loop placed entirely inside the body of another while loop. For every single iteration of the outer loop, the inner loop executes completely from start to finish.

The do...while Loop. The do...while loop is a variant of the while loop with one critical difference: its body is executed at least once before the condition is checked. It is a post-test loop, making it ideal for scenarios like password validation or menu prompts where execution must occur before evaluation.

3. SOFTWARE USED

- IDE: DEV-C++ IDE / VS Code
- Compiler: C++ Compiler

4. EXPERIMENTAL TASKS

Task 1: while Loop Counting

Write a C++ program that uses a while loop to display numbers from 1 to 10. Each number should be displayed on a separate line.

```
#include <iostream>
using namespace std;

int main() {
    int count = 1;

    // Execute as long as count is less than or equal to 10
    while (count <= 10) {
        cout << count << endl;
        count++; // Update variable
    }

    return 0;
}
```

Task 2: do...while Loop Counting

Repeat the previous task using a do...while loop to demonstrate post-test iteration, displaying numbers from 1 to 10.

```cpp
#include <iostream>
using namespace std;

int main() {
    int count = 1;

    // Execute body first, then check condition
    do {
        cout << count << endl;
        count++; // Update variable
    } while (count <= 10);

    return 0;
}
```

Task 3: Sum using a while Loop

Write a C++ program that prompts the user to enter a positive integer. Use a while loop to calculate the sum of all numbers from 1 to the entered integer, and display the final sum.

```cpp
#include <iostream>
using namespace std;

int main() {
    int limit, sum = 0, i = 1;

    cout << "Enter a positive integer: ";
    cin >> limit;

    // Calculate sum using while loop
    while (i <= limit) {
        sum += i;
        i++;
    }

    cout << "The sum from 1 to " << limit << " is: " << sum << endl;

    return 0;
}
```

5. CONCLUSION

This lab provided extensive practice with indefinite iteration using while and do...while loops. The primary distinction observed was that while loops evaluate their condition prior to execution, potentially running zero times, whereas do...while loops guarantee at least one execution. These structures are vital for developing dynamic programs that must respond repeatedly to varying user inputs and unknown bounds.